Calculus I (MAT 1001), L07

Syllabus

Fall, 2026, CUHK-SZ

Instructor:	Professor Clement Leung
Office:	Research A, Room 421
Office hours:	Tuesdays 4 – 6 pm
Email:	clementleung@cuhk.edu.cn
Lecture Time and Location	Mon & Wed 10:30 am-12:20 pm, Teaching Complex A402

Teaching Assistants:	Zhifei Xu	Liujia Ma	Yifei Chen
Offices:	Chengdao319A	TC 411b	Chengdao 319B
Office hours:	Tuesday 15:00-16:00	Monday 14:00-15:00	Monday 20:00-21:00
Emails:	224010188@link.cuhk.edu.cn	226010347@link.cuhk.edu.cn	226010434@link.cuhk.edu.cn
Classes:	T25	T27	T28

Prerequisites: High School Algebra, Trigonometry, Geometry, in particular, **Chapter 1 of the text book**

Text: *Thomas' Calculus (**14th Edition in SI Units**)*, George B. Thomas Jr., Maurice D. Weir, Joel R. Hass. Publisher: Pearson/高等教育出版社.

Reference: *Calculus: Early Transcendentals (8 or 9th Edition)*, James Stewart. Publisher: Brooks Cole.

Course Description: This course covers one-variable calculus; it emphasizes intuitive and conceptual understanding of theory of Calculus, computation skills, and nurtures the mentality and the ability to use Calculus to solve problems in other scientific disciplines.

Course Policies:

- The **semester grade** will be given according to performance in homework (10%), weekly quizzes (25%), mid-term (25%) and the final exam (40%).
- Homework will be assigned and graded weekly. Each homework assignment covers the sections taught in the lecture class in the same week and must be turned in in the tutorial class of the following week. No late homework will be accepted.
- There will be no make-up quizzes—three (3) worst quizzes will be dropped before calculating the semester grade.
- There will be one (**1) midterm exam**, lasting 2 hours. The tentative date of the midterm is **Saturday, Oct 24, from 9:00-11:00 am.**
- Missing the midterm or final exam without prior notification to and approval of me will automatically result in the "0" grade for the exam.

- Mid-term and final exam papers, as well as homework assignments, will be identical for all lecture sections; quizzes, however, will be created by your own course instructor and hence will differ by section; moreover, specific homework problems selected by the instructor and graded by the TAs may also vary by section. To ensure fairness, semester letter grades are assigned using a quota system: at the end of the semester, instructors receive a grade quota for their students based on their rankings in the common mid-term and final exams. Instructors then use the combined, weighted scores (homework, quizzes, mid-term, and final) to finalize letter grades within these quotas.

Course Contents:

Limit of a function and limit laws, one-sided limits, continuity, infinite limits, asymptotes of graphs
Tangents, rate of change and derivative at a point, derivative as a function, higher order derivatives, differentiation rules, derivative of trigonometric functions
Derivative of trigonometric functions(continued), chain rule, implicit differentiation, related rates
Linearization and differentials, extreme values of a function, mean value theorem, monotonicity, first derivative test
Concavity, second derivative test, curve sketching, applied optimization
Newton's method, area approximation by using left, right and mid sums, limits of these sums, definite integral
Fundamental theorem of calculus, antiderivatives and indefinite integrals, substitution method
Substitution method (continued), area between curves, volumes using cross-sections, arclength, areas of surfaces of revolution
Work and fluid forces, inverse functions and their derivatives, natural log, exponential functions
L'Hôpital's rule, inverse trigonometric functions, relative rates of growth, integration by parts
Trigonometric integrals, trigonometric substitutions, partial fractions method
Partial fractions method (continued), numerical integration: by rectangle, trapezoidal and Simpson's rules, improper integrals
First order ODE: solutions, slope fields, Euler's method, linear equations, separable equations
First order ODE: applications--"mixing problems", Malthusian model, Logistic model, autonomous equations, phase-line analysis

Homework Assignments:

2.1: #1, 7, 8, 19, 20
2.2: #1, 4(a)-(e), 9, 10, 18, 19, 23, 28, 32, 40, 44, 45, 54, 57, 65 (a), 75, 77, 80
2.3: skipped
2.4: #1, 5, 6, 17, 22, 30, 34, 36, 48, 49, 50
2.5: #7-10, 26, 27, 32, 41, 45, 49, 59, 61, 64
2.6: #2, 10, 11, 16, 21, 23, 31, 36, 39, 43, 63, 67, 73, 74, 84, 86, 103

3.1: #1, 2, 23, 33, 41, 47
3.2: #1, 20, 27-31, 34, 45,46, 48-50, 59, 60

3.3: #6, 11, 17, 20, 21, 29, 35, 39, 41, 46, 55-58
3.4: #3,7, 15, 21, 23, 25, 26
3.5: #3,12,15,17,26,32, 34， 47, 52,53, 58, 60, 61(a)
3.6: #5, 7, 10, 15, 29, 34, 41, 54, 67, 72, 75, 76, 84, 86,
3.7: #1, 12, 19, 31, 43
3.8: #7, 23, 27, 32, 42
3.9: #1, 6, 17, 18, 29, 37, 42, 51, 56

4.1：#1-6, 20, 36, 42, 45, 59, 60
4.2: #13, 15 a, (iii), 15b, 18, 20, 23, 27, 33, 60, 61, 63,65
4.3: #15-18, 23, 35, 55a, 67
4.4: #1,4,17,23,32,46, 75， 78, 95,97,106,107, 112
4.5: #1, 7, 12, 20a, 22, 28, 42, 52, 53, 55, 58, 71a
4.6: #1, 3, 9, 11, 12, 22
4.7: #1, 7, 12, 23, 32, 34, 52, 59, 63, 75, 85, 99, 103, 107

5.1：#1, 3，5, 7, 19
5.2: #8,12, 30, 32, 37, 39, 41, 42, 43, 44
5.3: #1, 3, 6, 8, 9, 17,20, 55, 71,73, 76, 77, 78, 81, 83
5.4: #1, 3, 4, 8, 9, 10, 12, 17, 22, 24, 27, 28, 30, 40, 44, 46, 61, 63, 68, 71, 73
5.5: #18, 20, 21, 22, 26, 31, 38, 42, 43, 46, 56, 62
5.6: #8, 11, 12, 21, 23, 25, 59, 63, 71, 72, 78, 82, 83, 84-87
6.1: #1, 10, 12, 14, 21, 25, 29, 31, 39, 43, 45, 53, 61
6.3: #1, 2, 9, 23, 30
6.4: #13, 18, 22, 23, 25, 28
6.5: #3, 9, 12, 16, 25, 26, 31, 38, 46
(6.2 and 6.6 skipped)

7.1: #1-6, 12, 14, 17, 25, 31, 38, 39, 44, 56, 59

7.2: #1, 4, 7, 10, 11, 23, 25, 36, 41, 44, 47, 55, 58, 67, 72, 77, 81, 89
7.3: #1, 8, 16, 18, 26, 28, 29, 33, 42, 52, 60, 63, 66, 69, 78, 79, 86, 89, 92, 95, 100, 103, 115, 122, 131, 133, 141, 146
7.4: delayed
7.5: #16, 20, 21, 26, 31, 38, 40, 41, 45, 51, 53, 59, 60, 61, 66, 67, 68,75, 85, 88
7.8: (taught before 7.6) #1，9-12, 20, 21(a), 22
7.6: #1, 3, 5, 7, 13, 16, 18, 22, 25, 33, 48, 51, 56, 67, 69, 73, 78, 91, 97, 98, 102, 103, 129,133
(7.7 skipped)

8.1: skipped
8.2: #2,6,12, 17, 23, 28, 29, 31, 32, 37, 43, 45, 50, 51, 53, 65, 67, 75
8.3: #3, 14, 15, 19, 20, 24, 27, 30, 33, 34, 35, 53, 54, 56, 64, 67
8.4: #2, 3, 5, 8, 9, 10, 15, 17, 21, 26, 36, 37, 61, 62
8.5: #6, 7, 13, 16, 20, 23, 27，31, 34, 35, 41
8.7: #8, 24, 25, 28, 29, 31
8.8: #4, 10, 14, 20, 23, 27, 32, 39, 43, 45, 56, 57, 60, 65, 68, 69, 71, 76, 80

16.1: #1-6, 18 (do 18 after learning 7.4 and 16.2)
7.4: #11, 12, 13, 17, 30, 34, 42
16.2: # 1, 4, 16, 17
16.3: # 13, 15
16.4: #1, 7, 10, 12, 14, 18 (for these problems, no need to study the sign of the second derivative of the solutions)